Appendix: Complete Hyperparameter Settings & Algorithm Steps

Table 1: Explicit values and definitions for all key hyperparameters in K-FedICGL.

Symbol	Hyperparameter Description	Value
1. Server-side: Invariant Causal Subgraph Alignment (Eq. 5-7)		
τ_c	Structural consensus threshold for initial filtering	0.5
γ	Penalty coefficient controlling sensitivity to cross-domain variance	1.0
τ_s	Active alignment score threshold for the final invariant mask $\mathbf{M}^{inv}$	0.5
2. Client-side: Local Optimization & Causal Contrastive Learning (Eq. 8-10)		
σ	Standard deviation of the injected Gaussian random noise $\epsilon \sim \mathcal{N}(0, \sigma^2\mathbf{I})$	0.1
τ	Temperature coefficient for the InfoNCE Causal Graph Contrastive Loss	0.5
β	Regularization strength for the contrastive penalty $\mathcal{L}_{cgc}$	0.1
3. Client-side: Knowledge-Driven Discovery (Eq. 3-4)		
λ	Regularization weight regulating the strength of the knowledge prior penalty	0.1
ρ	Initial penalty parameter for the Augmented Lagrangian	1.0

Algorithm 1 Training Process of the K-FedICGL Framework

Require: Clients $\mathcal{K}$, Local Spatio-temporal Data $\mathbf{X}^{(k)}$, Knowledge Priors $\mathbf{K}^{(k)}$, Comm. Rounds R, Local Epochs E.

Ensure: Global optimal predictor Θ and invariant causal skeleton $\mathbf{A}^{inv}$.

for round $r = 1, 2, \ldots, R$ **do**
  — Phase 1: Local Discovery (Client-side) —
  for each client $k \in \mathcal{K}$ **in parallel do**
    Compute prior penalty mask $\mathbf{M}^{(k)} = 1 - \mathbf{K}^{(k)}$.
    Optimize Eq. (4) via Augmented Lagrangian to generate local causal graph $\mathbf{A}^{(k)}$.
    Upload $\mathbf{A}^{(k)}$ to the Server.
  end for
  — Phase 2: Global Aggregation (Server-side) —
  Compute structural consensus matrix $\mathbf{C}$ with threshold τ_c (Eq. 5).
  Calculate cross-environment variance matrix $\mathbf{V}$ across all $\{\mathbf{A}^{(k)}\}$.
  Evaluate active alignment scores $\mathbf{S}_{i,j} = \mathbf{C}_{i,j} \cdot \exp(-\gamma \mathbf{V}_{i,j})$ (Eq. 6).
  Extract global invariant skeleton $\mathbf{A}^{inv} = \mathbb{I}(\mathbf{S} > \tau_s) \odot \bar{\mathbf{A}}$ (Eq. 7).
  Broadcast $\mathbf{A}^{inv}$ back to all active clients.
  — Phase 3: Local Optimization (Client-side) —
  for each client $k \in \mathcal{K}$ **in parallel do**
    Identify spurious environment edges $\mathbf{A}^{(k)}_{spur} = \max(0, \mathbf{A}^{(k)} - \mathbf{A}^{inv})$.
    for local epoch $e = 1, 2, \ldots, E$ **do**
      Intervene with $do(\epsilon)$ to generate counterfactual features $\tilde{\mathbf{Z}}^{(k)}_{cf}$ (Eq. 8).
      Compute InfoNCE causal graph contrastive loss $\mathcal{L}^{(k)}_{cgc}$ (Eq. 9).
      Update local parameters $\Theta^{(k)}$ by minimizing $\mathcal{L}^{(k)}_{local} = \mathcal{L}_{pred} + \beta \mathcal{L}^{(k)}_{cgc}$ (Eq. 10).
    end for
    Upload updated model parameters $\Theta^{(k)}$ to the Server.
  end for
  Server aggregates $\{\Theta^{(k)}\}$ via standard FedAvg to update global model Θ.
end for